Content

PHY 111: Introduction to Physics
PROBLEMS

Q1. An object of mass 4 *kg* moves round a circle of radius 6 *m* with a constant speed of 12 *m/s*. Calculate:
(i) the angular velocity,
(ii) the force towards the centre.

Q2. An object of mass 10 *kg* is whirled round a horizontal circle of radius 4 *m* by a revolving string inclined to the vertical. If the uniform speed of the object is 5 *m/s*, calculate:
(i) the tension in the string
(ii) the angle of inclination of the string to the vertical.

Q3. A racing-car of 1000 *kg* moves round a banked track at a constant speed of 108 *km/hr*. Assuming the total reaction at the wheels is normal to the track, and the horizontal radius of the track is 100 *m*, calculate the angle of inclination of the track to the horizontal and the reaction at the wheels.

Q4. Assuming that the mean density of the earth is 5500 kg/m^3, that the constant of gravitation is $6.7\text{x}10^{-11} Nm^2kg^{-2}$, and that the radius of the earth is 6400 *km*, find a value for the acceleration due to gravity at the earth's surface. Derive the formula used.

Q5. A copper wire, 200 *cm* long and 1.22 *mm* diameter, is fixed horizontally to two rigid supports 200 *cm* long. Find the mass in *grams* of the load which, when suspended at the mid-point of the wire, produces a sag of 2 *cm* at that point. Young's modulus for copper = $12.3\text{x}10^{10}\ Nm^{-2}$